

F1 — ISO 52016-1 against a matched EnergyPlus reference, baseline and corrected

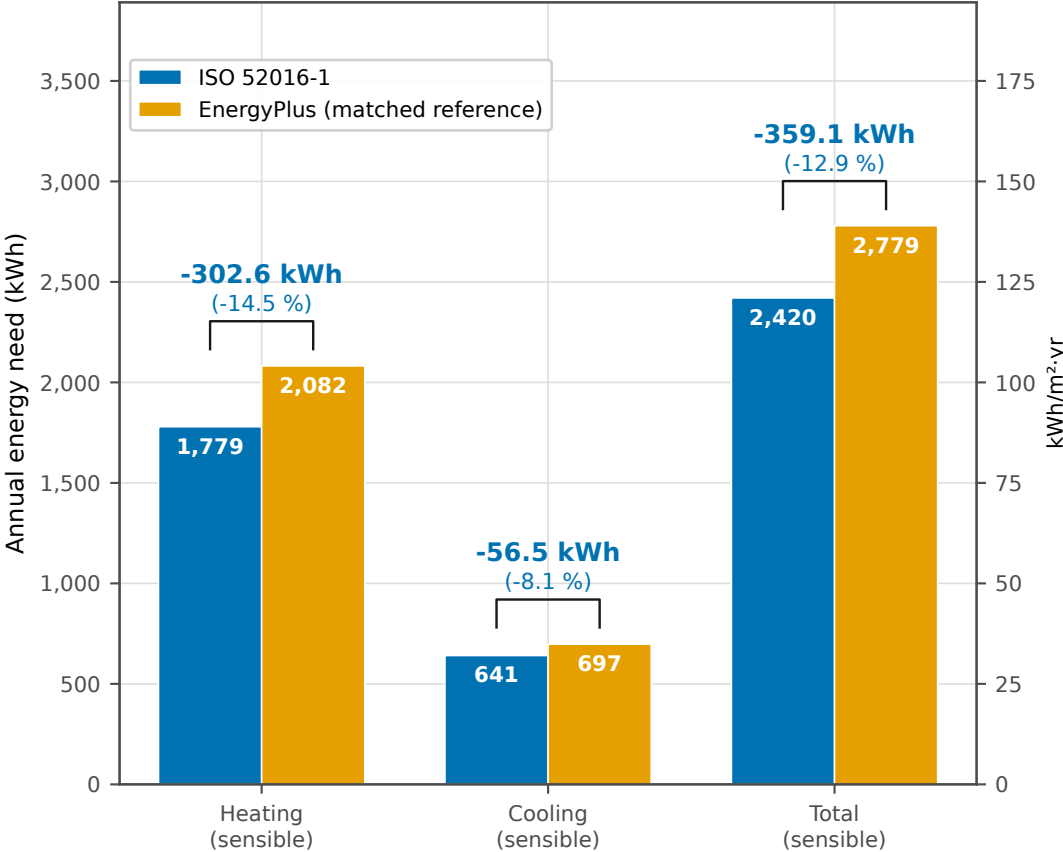
The ISO engine under-predicts on both engines and both components; every absolute gap narrows

Same weather (Melbourne-Essendon Fields TMYx 2011–2025); ideal loads; both engines sensible-only.

Each engine is compared against a reference matched to it, and both references carry the repairs to the four defects of Section 4.2. Differences are stated against the EnergyPlus reference.

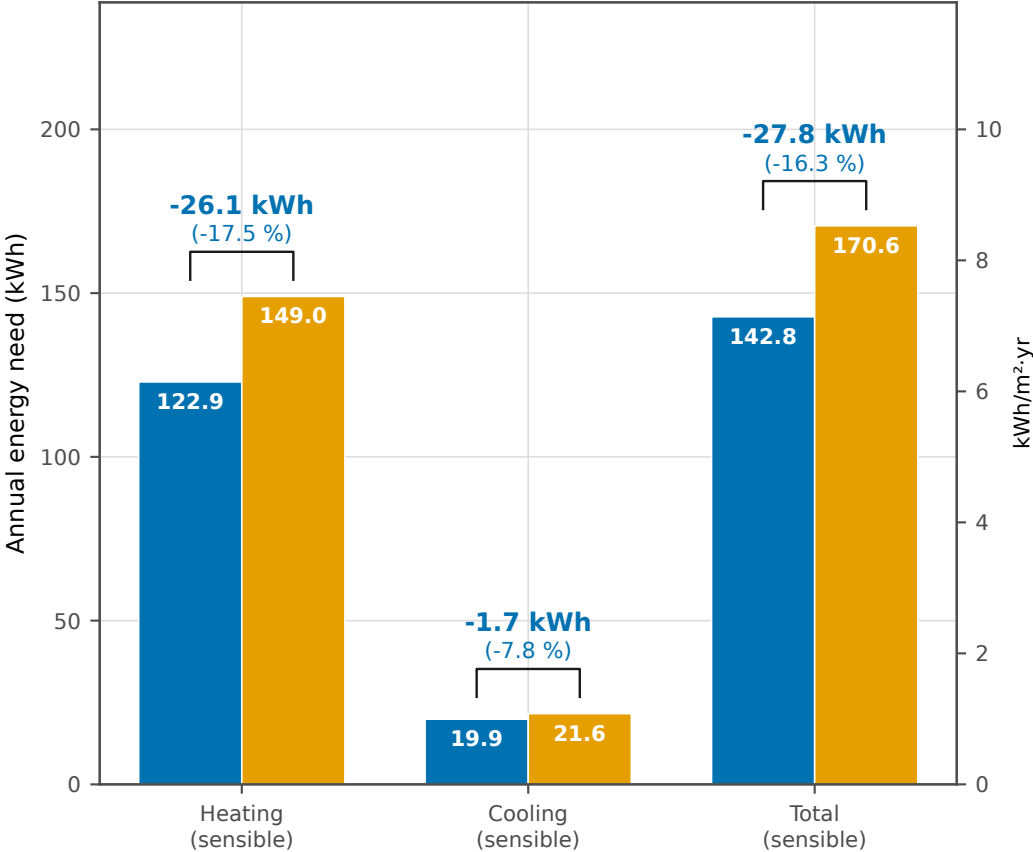
A Baseline engine

reference matched to it: unconditioned buffer neighbours, inflated internal gains, no infiltration



B Corrected engine

reference matched to it: neighbours at 20 °C, EN 16798-1 gains, infiltration at the adopted permeability, terrain-corrected wind



Read the absolute gap, not the percentage. Heating: -302.6 → -26.1 kWh, a 91 % narrowing, while the percentage moves the other way, -14.5 % → -17.5 %. Cooling: -56.5 → -1.7 kWh, a 97 % narrowing, at -8.1 % → -7.8 %.

The corrections cut the loads themselves by an order of magnitude, so a much smaller residual sits on a much smaller denominator. Relative error is not a stable measure as the load approaches zero; neither figure should be quoted without the other.